TEACHING PLAN

1. Course Name	High Voltage Engineering
2. Course Code	EEE 304
3. Credit Value	3
4. Academic Session	09-2023
5. Lecturer(s)	Ir Dr Mohd Faris Bin Abdullah
6. Tutor(s) (if any)	Nil
7. Course Learning Outcomes (CLO)	At the end of the course, students will be able to: 1. Analyze the methods of generations and measurements of high voltage AC, DC and impulse generator 2. Illustrate the various breakdown mechanism and applications of vacuum, liquid, solid and composite dielectrics 3. Determine the over-voltage phenomena, insulation coordination in power system, types of high voltage testing for electrical apparatus and non-destructive testing of materials
8. Sequence of planned lessons	**Topics**
Week 1	• Syllabus, OBE • Introduction to high voltage engineering
Week 2	• Conduction and breakdown in gases • Tutorial: Electric field stress
Week 3	• Conduction and breakdown in gases (cont.) • Conduction and breakdown in liquid • Tutorial: Electric field stress (cont.)
Week 4	• Conduction and breakdown in solid • Tutorial: Conduction and breakdown in gas
Week 5	• Conduction and breakdown in solid (cont.) • Application of insulating materials • Tutorial: Conduction and breakdown in gas (cont.)
Week 6	• Generation of high voltages and currents • Tutorial: Conduction and breakdown in solid • Lab: Impulse Voltage Generation and Electrical Breakdown Under Impulse Voltage experiment at UM high voltage lab on 27/10/2023 at 10.00am-12.00pm for group 1
Week 7	• Generation of high voltages and currents (cont.) • HV test lab • Lightning and risk assessment • Tutorial: Conduction and breakdown in solid (cont.) • Lab: Impulse Voltage Generation and Electrical Breakdown Under Impulse Voltage experiment at UM high voltage lab on 3/11/2023 at 10.00am-12.00pm for group 2
Week 8	• Lightning and risk assessment (cont.) • Tutorial: Insulation Breakdown Theory • Lab: Impulse Voltage Generation and Electrical Breakdown Under Impulse Voltage experiment at UM high voltage lab on 10/11/2023 at 10.00am-12.00pm for group 3
Week 9	• Mid-Term Examination – 15/11/2023 at 2.00-4.00pm room A5#G06 • Tutorial: Generation of high voltages and currents • Lab: Impulse Voltage Generation and Electrical Breakdown Under Impulse Voltage experiment at UM high voltage lab on 17/11/2023 at 10.00am-12.00pm for group 4

Week 10	• Lightning and risk assessment • Lightning protection • Tutorial: Generation of high voltages and currents (cont.) • Assignment: Lightning Risk Assessment
Week 11	• Electrical earthing • Safety in HV • Over-voltage phenomenon
Week 12	• Over-voltage phenomenon (cont.) • Insulation coordination • Tutorial: Over-voltage phenomenon
Week 13	• Insulation coordination (cont.) • Non-destructive insulation testing • Tutorial: Over-voltage phenomenon (cont.)
Week 14	• Non-destructive insulation testing (cont.) • High voltage measurement • Tutorial: Insulation coordination
Week 15	• High voltage measurement (cont.) • High voltage testing • Tutorial: Insulation coordination (cont.)

9. Required Materials (including equipment & technology)

No.	List of Practical Activities
1.	**Activities:** • Nil **Laboratory:** • Impulse voltage generation and electrical breakdown under impulse voltage impulse voltage experiment at UM high voltage lab **Materials/ Software used:** • Multisim, Pspice, Psim, Matlab/Simulink, EMTP-ATP and PSCAD

10. Assessments

Method of Assessment	Total
Continuous Assessment	**50%**
• Assignment	20%
• Laboratory	10%
• Midterm Examination	20%
Final Assessment	**50%**
• Final Examination	50%
TOTAL	**100%**

11. Assessments that align to Course Learning Outcomes (CLO)

Assessment	CLO 1	CLO2	CLO3	CLO4	CLO5
Continuous Assessment					
• Assignment			√		
• Laboratory	√				
• Midterm Examination		√			
Final Assessment					
• Final Examination	√	√	√		

12. Main Reference(s)	• E. Kuffel, W.S. Zaengl, J. Kuffel. (2016). 'High Voltage Engineering: Fundamentals', Newness.
13. Additional Reference(s)	• S. Naidu & V. Kamaraju, "High Voltage Engineering", McGrawHill, 2013. • Subir Ray. (2013). 'An Introduction to High Voltage Engineering', Prentice-Hall of India. • C.L. Wadhwa. (2010). 'High Voltage Engineering'. New Age International.